ROBOGENIX: AI-Powered Autonomous Mobile Robot

Project Report (Hardware Project)

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1. Abstract

ROBOGENIX is an AI-powered mobile robot designed to move autonomously, understand its surroundings and interact with users. The robot uses a camera for real-time object detection, ultrasonic sensors for obstacle detection, motors controlled by a microcontroller for movement, and a conversational AI (chatbot) to answer user queries. By combining autonomous navigation, computer vision and a chatbot in a single platform, ROBOGENIX demonstrates how robotics and artificial intelligence can work together to build intelligent, user-friendly systems.

2. Introduction

Robots are widely used in industries, healthcare, education and smart homes. However, many traditional robots depend on continuous human control and cannot understand their environment or communicate naturally with people. Recent progress in computer vision, embedded systems and conversational AI makes it possible to build robots that can sense, decide and respond on their own.

ROBOGENIX is a low-cost prototype that brings these technologies together. It can detect objects through a camera, avoid obstacles using sensors, change its direction automatically, and talk to users through a chatbot.

3. Problem Statement

Traditional robots often require human control and lack the ability to intelligently understand their surroundings and interact with users. There is a need for a robot that can move on its own, detect objects and obstacles, avoid collisions, and answer user questions, all in one integrated system.

4. Objectives

1. Assemble the robot using a camera, sensors, motors and a controller.
2. Implement object detection using computer vision.
3. Detect obstacles using ultrasonic sensors.
4. Program autonomous movement and direction changes.
5. Integrate a chatbot for answering user queries.
6. Test the robot's navigation, detection and interaction capabilities.

5. Scope of the Project

- The robot works as a prototype in indoor, controlled environments.
- It performs autonomous movement, object detection, obstacle avoidance and basic question answering.
- The system can later be extended to applications such as guidance, surveillance, education and assistance.

6. Hardware and Software Requirements

6.1 Hardware Components

Component	Purpose
Controller / Processing unit (e.g. Raspberry Pi / Arduino)	Brain of the robot; runs control logic and AI models
Camera module	Captures live video for object detection
Ultrasonic sensor (e.g. HC-SR04)	Measures distance to obstacles
DC motors with wheels	Provide movement
Motor driver (e.g. L298N)	Controls speed and direction of motors
Battery / power supply	Powers the controller, motors and sensors
Chassis	Frame that holds all components
Microphone and speaker (optional)	Voice input and output for the chatbot

(Update this table with the exact components you actually use.)

6.2 Software and Tools

- Python programming language
- OpenCV and a pre-trained object detection model (e.g. YOLO / MobileNet-SSD)
- Arduino IDE or embedded C/Python for sensor and motor control
- Chatbot / conversational AI module (API or rule-based)
- GitHub for documentation and version control

7. System Architecture

The system has four main blocks:

1. **Input Block:** camera, ultrasonic sensors, user query (text or voice).
2. **Processing Block:** controller that runs object detection, obstacle logic and the chatbot.
3. **Output / Actuation Block:** motor driver and motors for movement; screen or speaker for chatbot replies.
4. **Power Block:** battery supplying all modules.

Block diagram (flow of data):

Camera + Ultrasonic Sensors + User Query → Controller (Object Detection, Obstacle Logic, Chatbot) → Motor Driver → Motors / Speaker-Display

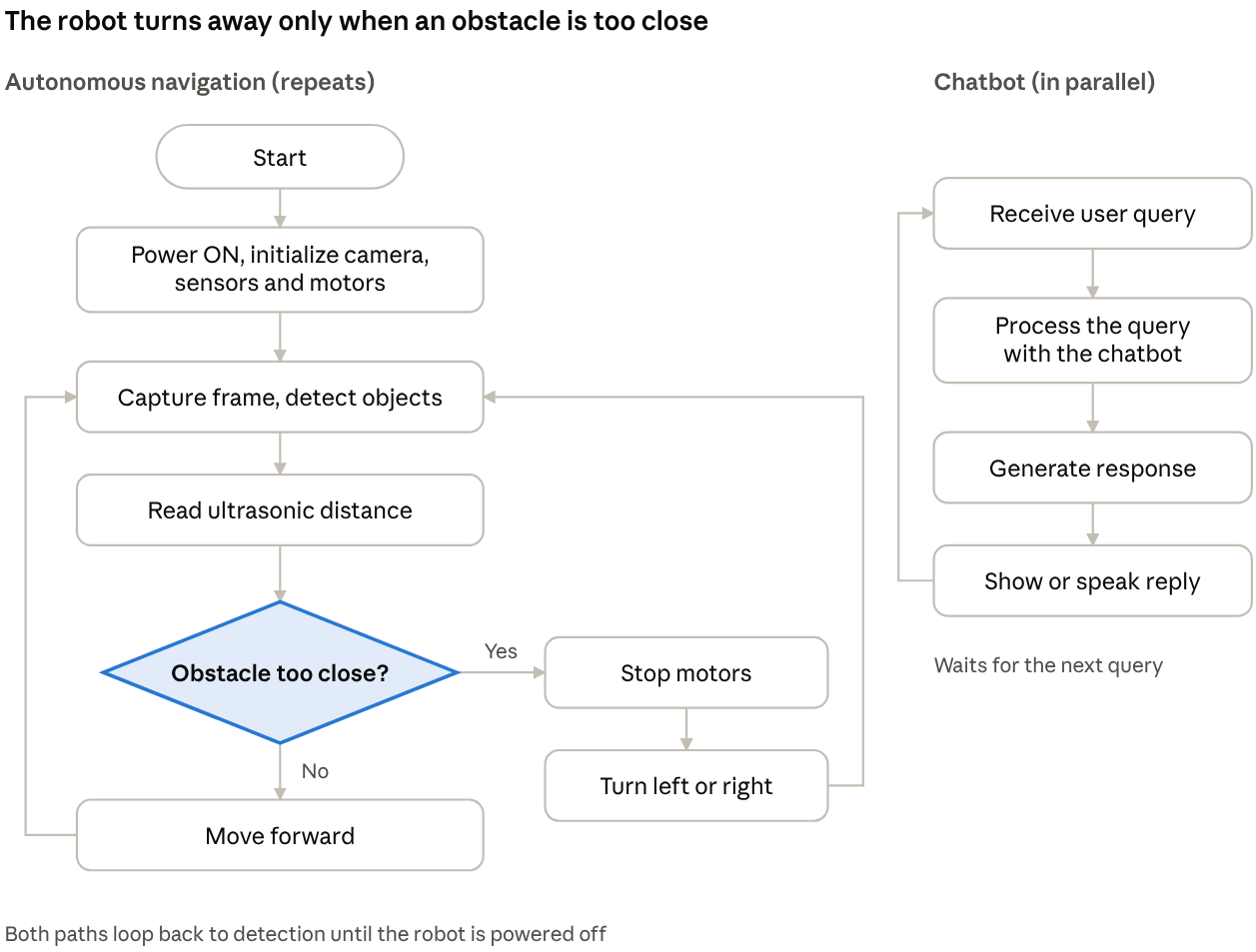
(Insert the block diagram image here, created in draw.io.)

8. Methodology and Working

1. **Assembly:** Mount the camera, sensors, motors and controller on the chassis and connect them through the motor driver and power supply.
2. **Object detection:** The camera streams video to the controller, where a computer vision model identifies objects in each frame.
3. **Obstacle detection:** The ultrasonic sensor continuously measures distance. If the distance falls below a set threshold, an obstacle is considered detected.
4. **Autonomous movement:** If the path is clear, the robot moves forward. If an obstacle is detected, the robot stops, turns to a new direction and continues moving.

5. **Chatbot interaction:** When a user asks a question, the chatbot processes it and gives an appropriate response.
6. **Testing:** Navigation, detection and interaction are tested separately and then together.

9. Flowchart



ROBOGENIX flowchart · navigation loop with 1 decision, plus chatbot loop

The robot reads the camera and ultrasonic sensor on every pass: if an obstacle is too close it stops and turns, otherwise it moves forward. The chatbot runs separately, answering each query and then waiting for the next.

Main navigation flow:

1. Start
2. Power ON the robot

3. Initialize camera, sensors and motors
4. Capture camera frame and run object detection
5. Read ultrasonic sensor distance
6. Decision: Is an obstacle detected within the threshold distance?
    - **Yes:** Stop motors → Change direction (turn left/right) → Go to step 4
    - **No:** Move forward → Go to step 4
7. Repeat until the robot is powered off
8. Stop

Chatbot flow (runs in parallel):

1. Wait for user query
2. Receive query (text/voice)
3. Process query using the chatbot
4. Generate response
5. Display / speak the response
6. Return to waiting

(Insert flowchart images here. Upload the PNG files to the GitHub repository under the `flowchart/` folder.)

10. Expected Outcomes

- Autonomous movement without continuous human control.
- Real-time object detection and obstacle avoidance.
- Automatic direction changes to prevent collisions.
- Intelligent responses to user queries through a chatbot.
- Integration of robotics, computer vision and AI into a single smart system.

11. Alignment with Sustainable Development Goals (SDGs)

- **SDG 9 – Industry, Innovation and Infrastructure:** Promotes innovation through AI, robotics and intelligent automation.
- **SDG 11 – Sustainable Cities and Communities:** Supports the development of smart environments through autonomous navigation and intelligent assistance.

- **SDG 4 – Quality Education:** Encourages hands-on learning in artificial intelligence, computer vision and robotics.

12. Applications

- Smart homes and offices (guidance and assistance)
- Educational and demonstration robots
- Surveillance and monitoring in indoor spaces
- Customer assistance in malls, hospitals or reception areas

13. Limitations

- Performance depends on lighting conditions and camera quality.
- Limited processing power on low-cost controllers can affect detection speed.
- Ultrasonic sensors may be less accurate on soft or angled surfaces.
- The chatbot may need an internet connection, depending on the implementation.

14. Future Scope

- Add mapping and path planning (SLAM) for smarter navigation.
- Add voice recognition and speech output for natural interaction.
- Use better sensors such as LiDAR for more accurate obstacle detection.
- Connect to a mobile app for remote monitoring and control.
- Train a custom object detection model for specific tasks.

15. Conclusion

ROBOGENIX shows how robotics, computer vision and conversational AI can be combined in one intelligent system. The robot can move autonomously, detect objects, avoid obstacles by changing direction, and respond to user queries. The project provides hands-on experience in AI and robotics and is a base for building more advanced intelligent robots in the future.

16. References

1. OpenCV documentation – https://docs.opencv.org
2. Raspberry Pi / Arduino official documentation
3. HC-SR04 ultrasonic sensor datasheet
4. United Nations Sustainable Development Goals – https://sdgs.un.org/goals

(Add any other books, papers or tutorials you use.)